

NANYANG TECHNOLOGICAL UNIVERSITY
SINGAPORE

College of Computing and Data Science

SC2006 – Software Engineering

Lab 1 Deliverables

SCSJ Group 3

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Functional & Non-Functional Requirements

Target Users

Commuters, students, and residents in Singapore who want to check real-time crowd levels at MRT/LRT stations, malls, and attractions before travelling there, so they can avoid overcrowded venues or choose a quieter alternative nearby.

A. Functional Requirements

1. Crowdy shall continuously aggregate real-time crowd data from official government APIs into a single unified format.
    a. Crowdy shall fetch real-time MRT/LRT station crowd levels from LTA DataMall's Station Crowd Density API.
    b. Crowdy shall fetch real-time carpark lot availability from LTA DataMall's Carpark Availability API for supported malls and attractions.
    c. Crowdy shall estimate a crowd percentage for each carpark-linked venue by comparing its current available lots against the highest availability historically observed for that carpark.
    d. Crowdy shall persist carpark crowd history to storage so that estimates are not lost when the system restarts.
    e. Crowdy shall automatically reposition venues that share near-identical coordinates so that each remains individually visible and selectable on the map.
    f. If a data source fails to respond, Crowdy shall continue serving the last successfully retrieved data for that source rather than showing no data.

2. Crowdy shall allow Users to view nearby venues and their live crowd levels.
    a. Crowdy shall display the User's current location on an interactive map, based on device GPS.
    b. Crowdy shall display colour-coded, category-labelled pins on the map representing different venue types (e.g., MRT, LRT, Mall, Attraction).
    c. Crowdy shall display a list of nearby venues below the map, sorted by distance, each showing its crowd level.
    d. As the User pans or zooms the map, Crowdy shall update the list to show only venues within the currently visible map area.
    e. Crowdy shall allow the User to recentre the map on their current location after panning or zooming away from it.

3. Crowdy shall allow Users to view detailed crowd information for a specific venue.
    a. Crowdy shall display the venue's name, category, address, current crowd level, and crowd percentage.
    b. Crowdy shall display when the venue's crowd data was last updated and which source it came from.
    c. Crowdy shall recommend lower-crowd alternative venues of the same category located near the selected venue.

4. Crowdy shall reduce visual clutter on the map by grouping nearby venues into a single clustered marker when the map is zoomed out.
    a. Crowdy shall expand a clustered marker back into its individual venues as the User zooms in far enough that they no longer overlap.
    b. Crowdy shall zoom the map in when the User taps a clustered marker.

5. Crowdy shall allow Users to sign up for an account to store their favourite venues.
    a. Crowdy shall allow Users to mark or unmark a venue as a favourite from its detail screen or list, whether or not they are logged in.
    b. If the User is logged in, Crowdy shall store favourites against their account.
    c. If the User is not logged in, Crowdy shall store favourites locally on the device.
    d. When a User with local favourites logs in or signs up, Crowdy shall merge those favourites into their account.
    e. Crowdy shall display a dedicated view listing the User's favourited venues, each showing its current crowd level.

B. Non-Functional Requirements

Category	Requirement
Usability	The map and list must remain responsive while panning/zooming, with no perceptible lag re-filtering the visible venue list.
Portability	The application must run on iOS, Android, and Web from a single shared codebase, with feature parity except where a platform genuinely cannot support a capability (e.g., no native map rendering on Web).
Performance	Station crowd data must be cached and refreshed no more than once every 5 minutes; carpark data no more than once every 1 minute – matching each source's own real-world update frequency, to avoid unnecessary load on external APIs.
Reliability	If an external data source is unreachable, the system must continue serving previously cached data for that source rather than failing the entire request.
Supportability	The venue data layer must be swappable (e.g., mock data, live API, future data sources) without requiring changes to the screens that display it.
Supportability	The crowd-history persistence layer must be replaceable with a standard SQL/NoSQL database without changing the API contract above it.
Supportability	Account and favourite data shall be stored in a MySQL database, consistent with the tooling supported in this course's lab environment.
Security	User passwords must be stored using a strong one-way hash (e.g., bcrypt); plain-text password storage is not permitted, regardless of the number of accounts in use.

Data Dictionary

Entity	Attribute	Data Type	Description
Venue	id	String	Unique identifier for the venue
	name	String	Display name of the venue
	category	Enum	MRT, LRT, Malls, Attraction
	address	String	Human-readable address or location description
	lat	Float	Latitude coordinate (WGS84)
	lng	Float	Longitude coordinate (WGS84)
	crowdPercent	Integer	Estimated crowd level as a percentage (0–100)
	crowdLevel	Enum	Low, Moderate, High, or Very High
	source	Enum	LTA, GooglePopularTimes, or Mock
	lastUpdated	DateTime	When this venue's crowd data was last refreshed
	distanceKm	Float	(optional) Distance from the User or map viewport centre, computed at query time
Station	code	String	Official LTA station code (e.g., "NS25")
	id	String	Internal identifier derived from name and mode
	name	String	Official station name
	mode	Enum	MRT or LRT
	lat, lng	Float	Coordinates derived from LTA's official geospatial dataset
	address	String	Display address for the station
CarparkHistory	carParkId	String	LTA's identifier for the carpark
	observedMaxLots	Integer	Highest available-lots value ever recorded; used as an estimated capacity baseline since LTA gives no official total
MapRegion	latitude, longitude	Float	Centre point of the currently visible map area
	latitudeDelta, longitudeDelta	Float	Span (zoom level) of the currently visible map area
User	location	{lat, lng}	The User's current device GPS position; used to sort/filter venues, not persisted beyond the session
Account	accountId	String	Unique identifier for the account
	email	String	Email address used to sign up / log in
	passwordHash	String	Securely hashed password (never stored in plain text)
	createdAt	DateTime	When the account was created

Favourite	favouriteId	String	Unique identifier for the favourite entry
	venueId	String	The venue being favourited
	accountId, deviceId	String	Exactly one is set: accountId if the User was logged in when saved, deviceId otherwise
	createdAt	DateTime	When the venue was favourited

Relationships

- A Venue has zero or one associated Station (when category is MRT or LRT) – 1:1
- A Venue has zero or one associated CarparkHistory record (when category is Mall/Attraction sourced from LTA carpark data) – 1:1
- An Account has zero or many Favourites – 1: many
- Each Favourite references exactly one Venue, and belongs to either exactly one Account (if saved while logged in) or one device (if saved as a guest), never both – many:1
- When a device's local Favourites are merged into an Account at login/sign-up, those records are re-pointed from deviceId to accountId
- A User's location is used transiently to query/filter Venues; it is not a persisted relationship between entities
- The System (via AggregateCrowdData) produces and refreshes the entire collection of Venue records; this is a collection-level relationship, not a link between individual records

Initial Use Case Model

1. Use Case Diagram

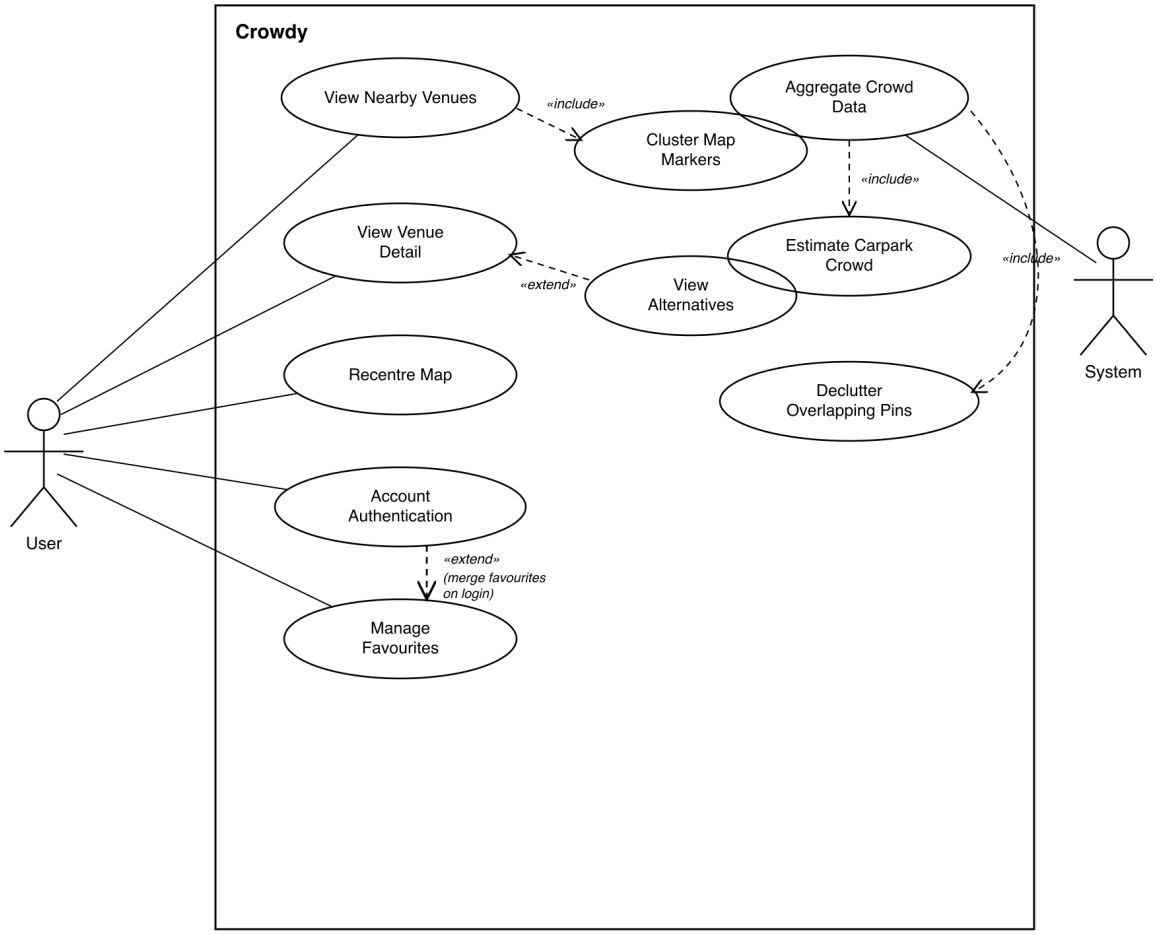


Fig. 1 – View Alternatives only fires when a lower-crowd option exists nearby, so it extends rather than always running as part of View Venue Detail. Cluster Map Markers is always invoked as part of View Nearby Venues (grouping is automatic display behaviour, not a deliberate User action), whereas Recentre Map is a discrete action the User initiates. On the System side, Aggregate Crowd Data always performs both of its included steps every refresh cycle. Manage Favourites works whether or not the User is logged in. Account Authentication extends Manage Favourites: logging in mid-flow triggers merging any local guest favourites into the account, an optional call the «extend» relationship is built to represent.

2. Use Case Descriptions

ViewNearbyVenues	
Actor	User
Description	Allows the User to view a map and list of nearby venues along with their current crowd levels.
Preconditions	The User has granted location permission (or the system falls back to a default location).
Postconditions	The map and list display venues within the User's vicinity, each showing a crowd level.
Priority	High
Frequency of Use	High
Flow of Events	**1.** User opens the app. **2.** System requests the User's current location. **3.** System retrieves nearby venues and crowd levels via AggregateCrowdData. **4.** System displays venues as colour-coded pins on a map centred on the User. **5.** System displays the same venues as a distance-sorted list below the map. **6.** If the User pans/zooms the map, the system updates the list to only show venues within the visible area.
Alternative Flows	If location permission is denied, the system centres the map on a default location (Marina Bay) instead.
Exceptions	If no venues are available (e.g., network failure), the system displays an empty-state message.
Includes	AggregateCrowdData, ClusterMapMarkers
Assumptions	The device has GPS-capable location services.
Notes/Issues	NIL

ViewVenueDetail	
Actor	User
Description	Allows the User to select a venue and view its detailed crowd information.
Preconditions	The User is viewing the nearby venues list or map (ViewNearbyVenues has been performed).
Postconditions	Venue detail information is displayed to the User.

Priority	High
Frequency of Use	High
Flow of Events	**1.** User selects a venue from the map or list. **2.** System retrieves the venue's full details (name, category, address, crowd %, crowd level, source, last updated). **3.** System displays the venue detail screen. **4.** If a lower-crowd alternative exists nearby, the system invokes ViewAlternatives.
Alternative Flows	NIL
Exceptions	If the venue's data is no longer available, the system displays a "not found" message.
Includes	None (extended by ViewAlternatives)
Assumptions	NIL
Notes/Issues	NIL

ViewAlternatives	
Actor	User
Description	Recommends lower-crowd venues of the same category located near the currently viewed venue. Extends ViewVenueDetail.
Preconditions	The User is viewing a venue's detail screen.
Postconditions	A list of lower-crowd alternative venues, if any, is displayed.
Priority	Medium
Frequency of Use	Medium
Flow of Events	**1.** System identifies venues of the same category as the current venue. **2.** System filters to venues within 3km. **3.** System filters to venues with a lower crowd percentage. **4.** System sorts remaining venues by distance. **5.** System displays up to three alternatives. **6.** If the User selects one, the system invokes ViewVenueDetail for it.
Alternative Flows	If no qualifying alternative exists, nothing is displayed in this section.
Exceptions	NIL
Includes	NIL
Assumptions	NIL
Notes/Issues	NIL

RecentreMap	
Actor	User
Description	Allows the User to recentre the map on their current location after panning or zooming away from it.
Preconditions	The User is viewing the map (ViewNearbyVenues) and has moved the view away from their current location.
Postconditions	The map recentres on the User's current GPS location at the default zoom level.
Priority	Low
Frequency of Use	Medium
Flow of Events	**1.** User taps the recentre control on the map screen.

	2. System retrieves the User's current GPS location. **3.** System animates the map back to centre on that location.
Alternative Flows	NIL
Exceptions	If location permission has been revoked, the system prompts the User to re-enable it.
Includes	NIL
Assumptions	NIL
Notes/Issues	NIL

ClusterMapMarkers	
Actor	User (included by ViewNearbyVenues)
Description	Groups map markers that are close together at the current zoom level into a single clustered marker showing a count, to reduce visual clutter; expands back into individual markers as the User zooms in.
Preconditions	The map is displaying two or more venues within a small area at the current zoom level.
Postconditions	Overlapping/nearby markers are shown as a single grouped marker with a count badge.
Priority	Medium
Frequency of Use	High
Flow of Events	**1.** System evaluates the current zoom level and on-screen positions of all visible venue markers. **2.** System groups markers that would visually overlap into a single cluster marker showing the count of grouped venues. **3.** If the User zooms in far enough that a cluster's markers no longer overlap, the system expands the cluster back into individual markers. **4.** If the User taps a cluster marker, the system zooms the map in on that cluster's area.
Alternative Flows	NIL
Exceptions	NIL
Includes	NIL
Assumptions	NIL
Notes/Issues	Distinct from DeclutterOverlappingPins, which repositions venues sharing near-identical real-world coordinates at the data layer. Clustering is a display-time grouping based on current zoom level, independent of exact coordinates.

AggregateCrowdData	
Actor	System
Description	Automatically and periodically collects crowd data from external sources and merges it into one unified dataset.
Preconditions	None (runs automatically on a schedule/cache expiry).
Postconditions	An up-to-date, merged list of venues with current crowd levels is available to serve to Users.
Priority	High
Frequency of Use	High (every 1–5 minutes depending on source)
Flow of Events	**1.** System checks whether cached station data has expired (5 min).

	2. If expired, fetches real-time station crowd levels from LTA per line. **3.** System checks whether cached carpark data has expired (1 min). **4.** If expired, fetches carpark availability from LTA, invoking EstimateCarparkCrowd per carpark. **5.** System merges station, carpark, and placeholder venue data into one list. **6.** System invokes DeclutterOverlappingPins on the merged list before serving it.
Alternative Flows	If a data source fails to respond, the system continues serving its last successfully cached result.
Exceptions	If a source errors for a subset of records (e.g., one train line), the system skips only that subset and continues.
Includes	EstimateCarparkCrowd, DeclutterOverlappingPins
Assumptions	NIL
Notes/Issues	Requires a valid LTA DataMall API account key.

EstimateCarparkCrowd	
Actor	System
Description	Estimates a crowd percentage for a carpark-linked venue, since LTA provides available lots but not total capacity.
Preconditions	Live available-lot data for the carpark has been retrieved.
Postconditions	A crowd percentage and level are computed and persisted for the carpark.
Priority	High
Frequency of Use	High
Flow of Events	**1.** System retrieves the carpark's previously recorded highest observed available-lot count. **2.** System compares this against the current count; if higher, the current count becomes the new maximum. **3.** System computes crowd percentage as the proportion of the maximum currently unavailable. **4.** System maps the percentage to a crowd level. **5.** System writes the updated maximum back to persisted storage.
Alternative Flows	NIL
Exceptions	If no prior history exists (first-ever reading), the system initialises the maximum to the current reading, yielding 0%.
Includes	NIL
Assumptions	A carpark's true capacity is reasonably approximated by the highest availability ever observed for it.
Notes/Issues	Requires persisted storage that survives a system restart. This is an estimate, not an official capacity figure; accuracy improves the longer the system runs.

DeclutterOverlappingPins	
Actor	System
Description	Repositions venues sharing a near-identical map location so each remains individually visible and selectable.
Preconditions	A merged list of venues with coordinates is available.

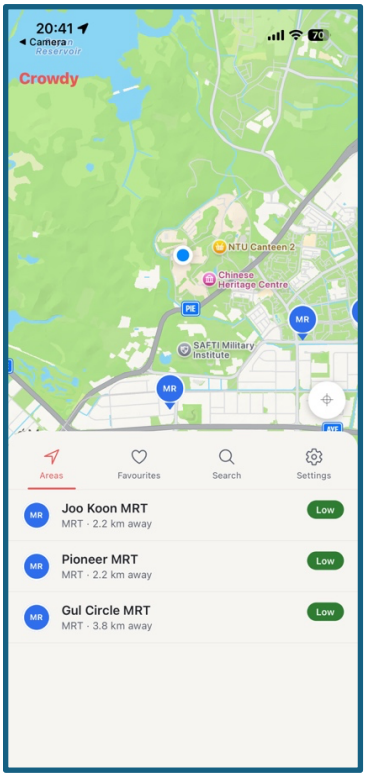
Postconditions	No two venues occupy the same small map area; each is independently tappable.
Priority	Medium
Frequency of Use	High
Flow of Events	**1.** System groups venues by rounding coordinates to a small grid cell. **2.** For each group with more than one venue, the system computes the shared centre point. **3.** System arranges the group's venues in a small circle around that centre point. **4.** System returns the adjusted coordinates for the map and list.
Alternative Flows	NIL
Exceptions	NIL
Includes	NIL
Assumptions	NIL
Notes/Issues	NIL

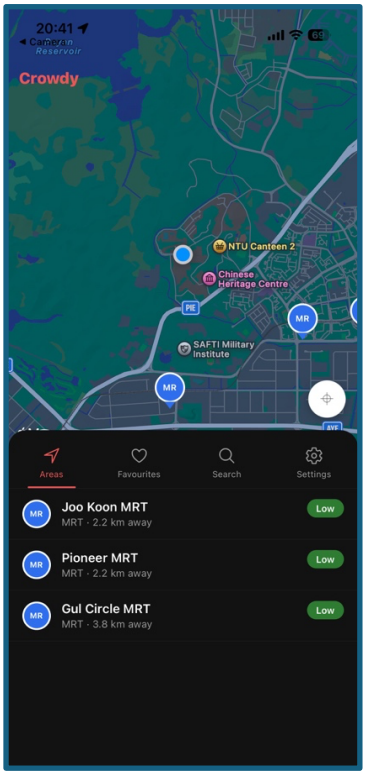
AccountAuthentication	
Actor	User
Description	Allows the User to sign up for a new account or log into an existing one, so favourites can be saved to their account and synced across devices.
Preconditions	None.
Postconditions	The User is authenticated and their session is associated with their account.
Priority	Low
Frequency of Use	Low
Flow of Events	**1.** User selects "Sign Up" or "Log In" from the Favourites view. **2.** If signing up, the User enters an email and password; the system validates the input and creates a new account with a securely hashed password. **3.** If logging in, the User enters their email and password; the system verifies the credentials against the stored account. **4.** If the User had local favourites saved as a guest, the system invokes ManageFavourites to merge them into the newly authenticated account. **5.** System confirms authentication and returns the User to the Favourites view.
Alternative Flows	If login credentials are incorrect, the system displays an error and prompts the User to retry.
Exceptions	If the email is already registered during sign-up, the system prompts the User to log in instead.
Includes	NIL
Assumptions	NIL
Notes/Issues	Requires a MySQL database to store account records; passwords must be securely hashed (e.g., bcrypt), never stored in plain text.

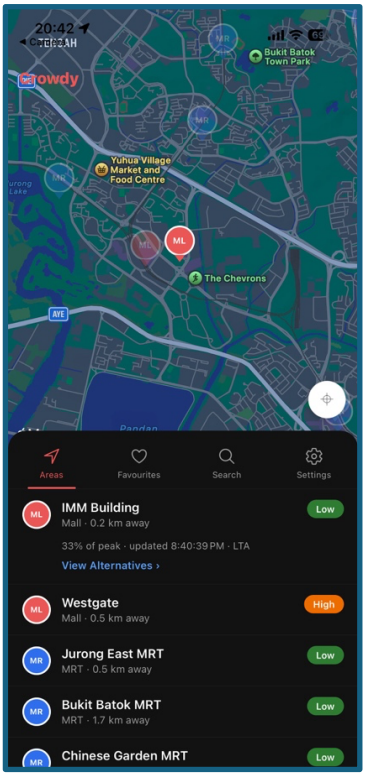
ManageFavourites	
Actor	User

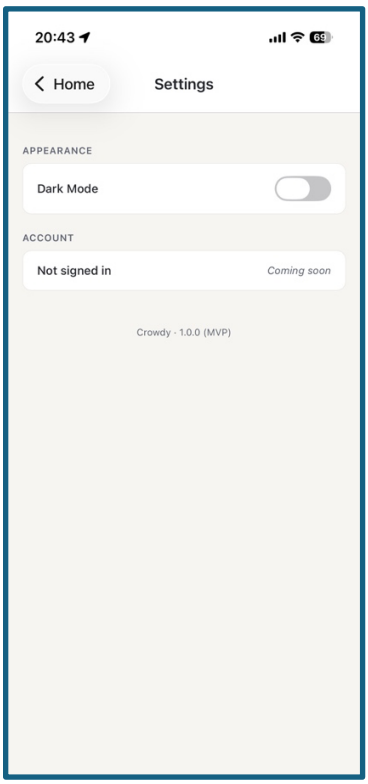
Description	Allows the User to save or remove venues as favourites, stored against their account if logged in or locally on the device otherwise, and view them in a dedicated list.
Preconditions	The User is viewing a venue's detail screen or the nearby venues list.
Postconditions	The venue is added to or removed from the User's favourites (account-linked or local, depending on login state).
Priority	Medium
Frequency of Use	Medium
Flow of Events	**1.** User taps the favourite icon on a venue's detail screen or list row. **2.** If the User is logged in, the system saves or removes the favourite against their account. **3.** If the User is not logged in, the system saves or removes the favourite in local device storage. **4.** System updates the favourite icon to reflect the new state. **5.** User may navigate to a dedicated Favourites view listing all saved venues, each with its current live crowd level. **6.** If the User subsequently logs in or signs up via AccountAuthentication, the system merges any local favourites into their account.
Alternative Flows	NIL
Exceptions	If a previously favourited venue no longer exists in the data source, the system removes it from the favourites list automatically.
Includes	None (invokes AccountAuthentication only if the User chooses to log in or sign up)
Assumptions	NIL
Notes/Issues	Requires on-device storage for guests, and a MySQL database for account-linked favourites.

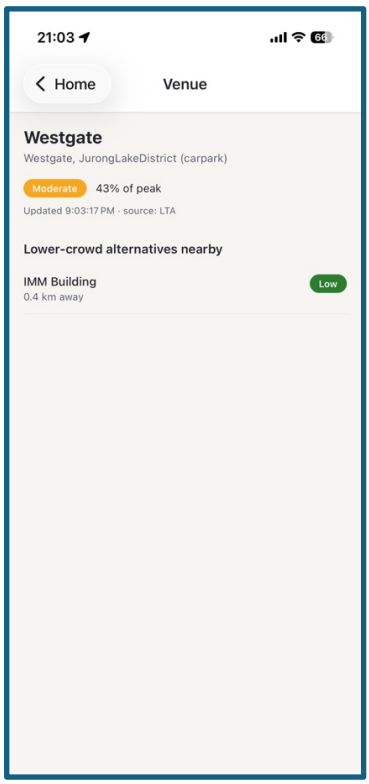
UI Mockups

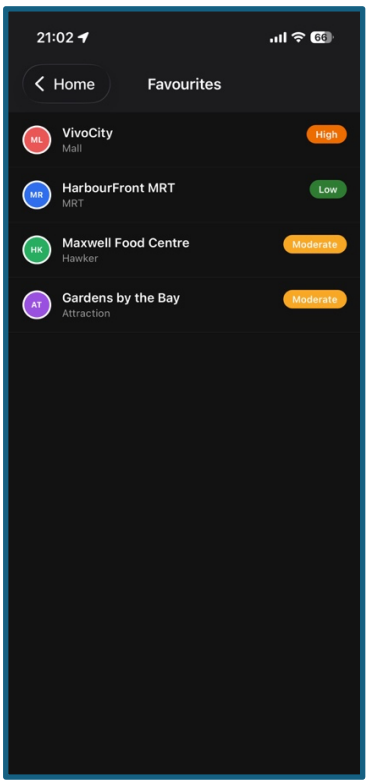

20:41
Crowdy
NTU Canteen 2
Chinese Heritage Centre
SAFTI Military Institute
Areas
Favourites
Search
Settings
Joo Koon MRT
MRT · 2.2 km away
Low
Pioneer MRT
MRT · 2.2 km away
Low
Gul Circle MRT
MRT · 3.8 km away
Low


20:41
Crowdy
NTU Canteen 2
Chinese Heritage Centre
SAFTI Military Institute
Areas
Favourites
Search
Settings
Joo Koon MRT
MRT · 2.2 km away
Low
Pioneer MRT
MRT · 2.2 km away
Low
Gul Circle MRT
MRT · 3.8 km away
Low


20:42
Crowdy
Bukit Batok Town Park
Yuhua Village Market and Food Centre
The Chevrons
Areas
Favourites
Search
Settings
IMM Building
Mall · 0.2 km away
Low
33% of peak · updated 8:40:39 PM · LTA
View Alternatives ›
Westgate
Mall · 0.5 km away
High
Jurong East MRT
MRT · 0.5 km away
Low
Bukit Batok MRT
MRT · 1.7 km away
Low
Chinese Garden MRT
Low


20:43
Home
Settings
APPEARANCE
Dark Mode
ACCOUNT
Not signed in
Coming soon
Crowdy · 1.0.0 (MVP)


21:03
Home
Venue
Westgate
Westgate, JurongLakeDistrict (carpark)
Moderate
43% of peak
Updated 9:03:17 PM · source: LTA
Lower-crowd alternatives nearby
IMM Building
0.4 km away
Low


21:02
Home
Favourites
VivoCity
Mall
High
HarbourFront MRT
MRT
Low
Maxwell Food Centre
Hawker
Moderate
Gardens by the Bay
Attraction
Moderate

AI Critique Report

(1) Prompt & AI Response

Prompt: "Critique the attached use case diagram image, with respect to how well it matches the attached requirements file."

Response: The diagram traces well to the requirements overall (FR1→Aggregate Crowd Data, FR2→View Nearby Venues, FR3→View Venue Detail, FR5→Manage Favourites/Account Authentication). One gap: FR5d says logging in merges a guest's local favourites into their account — a conditional call, which is exactly what «extend» is for. The diagram instead shows Account Authentication and Manage Favourites as two unconnected ovals, losing that relationship in the diagram itself.

(2) Quoted Aspect

"FR5d... is exactly what «extend» is for... two unconnected ovals... loses that relationship."

(3) Justification

We picked this because it corrected our own misunderstanding — we'd avoided «extend» thinking it meant "runs every time," when it's actually meant for conditional calls like this one. We've since added the «extend» arrow to Fig. 1.